Homework 3: Worked Solutions

Asset Returns and Geometric Brownian Motion

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Purpose and current status. This is the requested question-by-question worked solution. T1–T8 and A1–A3 include derivations, implemented code and actual final-count outputs. I1 contains two independently prompted DeepSeek audit sessions with the same-provider limitation disclosed, plus genuine follow-ups. The supplied I2 original is preserved and actual repairs/tests are documented. R1 contains source-based notes on two different books, Tsay and Hull, with independent reconstructions, checked examples, and a cross-book comparison; the AI-assisted source consultation and the edition handling are disclosed in the notes.

AI disclosure. The student reported having no earlier independent attempt, pre-AI debugging record or reading record. These solutions are AI-assisted; no independent student authorship, prior chronology, personal reading, or invented conversations are asserted. Mathematical/numerical cross-checks were performed in this session, and actual outputs, original files and model replies are retained. The student must study the arguments and complete the source reading personally before making any corresponding submission declaration.

Conventions. Time is in years; $S_0 > 0$, μ is physical annual arithmetic drift, and $\sigma \geq 0$ is annual volatility. Sample variances use divisor $n - 1$ unless an explicitly labelled population/moment calculation says otherwise. Master seed 21603 uses separate PCG64/SeedSequence child streams: A1 permutation $(1,)$, A2 $(2,)$, A3 $(3,)$, stress $(4,)$. The I2 standalone API uses its documented seed argument separately.

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1 Part I: Theory

1.1 T1. Weak-form efficiency and empirical evidence

(a) Let $\mathcal{F}_t^{\text{hist}}$ contain the prices and trading volumes available by time t. Weak-form efficiency means that this information is already reflected in current prices: a feasible strategy based only on $\mathcal{F}_t^{\text{hist}}$ should not systematically earn positive expected abnormal returns. "Abnormal" requires a specified equilibrium expected-return model or an appropriate passive benchmark, with comparable market and other risk exposures. Performance should be evaluated after commissions, the bid–ask spread, financing, borrowing fees, and other implementation costs. A return above the risk-free rate is not by itself abnormal; it can compensate for risk. The claim concerns persistent expected performance, not every realized trade or a guaranteed outcome in every sample.

(b) A non-Gaussian marginal return distribution is compatible with informational efficiency; neither Gaussianity nor independent, identically distributed returns is part of its definition. A profitable realized trade may be luck or compensation for bearing risk. A small sample autocorrelation measures only one form of linear dependence and is uncertain in finite samples. It neither proves independence nor excludes conditional variance dependence, nonlinear predictability, or an economically useful strategy after costs. Conversely, detectable dependence need not imply exploitable risk-adjusted profits. An empirical efficiency test also tests the chosen expected-return benchmark.

1.2 T2. Five-sigma events and observation frequency

(a) Under independent, identical Bernoulli trials with event probability $p_5 = 5.733 \times 10^{-7}$, the waiting time N in observations has

$$\mathbb{P}(N = k) = (1 - p_5)^{k-1} p_5, \qquad \mathbb{E}N = \frac{1}{p_5}.$$

With $252q$ observations per year, the expected waiting time in years is

$$\mathbb{E}\tau = \frac{1}{252 q p_5}.$$

The rounded statement of 7,000 years implies

$$q = \frac{1}{7000 \cdot 252 \cdot 5.733 \times 10^{-7}} = 0.988825\ldots,$$

which is approximately one observation per trading day. Exactly one daily observation gives $\mathbb{E}\tau = 6921.776$ years. Increasing the number of genuinely comparable independent observations per day reduces this waiting time proportionally; overlapping or dependent observations do not automatically create independent trials.

(b) Five years contain 1260 daily trials, so

$$\mathbb{P}(\text{at least one event}) = 1 - (1 - p_5)^{1260} = 0.0007220974 \approx 0.07221\%.$$

An empirical frequency of one event every three to four years is much larger than this fixed i.i.d.-Gaussian benchmark predicts. It suggests checking the benchmark, the volatility estimate, the sampling frequency, and possible regime changes. It does not identify a particular alternative distribution, establish a mathematical tail class, prove independence or dependence, or by itself reject weak-form efficiency. A short record also estimates rare-event frequencies imprecisely.

1.3 T3. Trading assumptions and feasible strategies

No transaction costs. In practice commissions, exchange fees, price impact, and financing expenses reduce realized returns. If a strategy turns over V dollars and incurs an effective proportional cost c, its approximate net trading gain is its gross gain minus cV. Frequent rebalancing can therefore erase a small theoretical advantage. This mainly changes execution cost and profitability; replacing discrete costly rebalancing with a frictionless hedge also introduces model risk when assessing its real performance.

Unrestricted short selling. A desired short may be unavailable, limited by a borrowing constraint, subject to a borrowing fee, recalled, or affected by a short-sale restriction. An unavailable borrow changes the feasible set: a mathematically specified position cannot necessarily be entered or maintained. For an available borrow, fees change its cost. An analyst who assumes unlimited stable borrowing can overstate arbitrage opportunities and understate hedging risk; a price discrepancy may persist because the proposed trade cannot be implemented at the modeled scale.

1.4 T4. Return aggregation and distributional diagnostics

(a) For positive prices,

$$1 + R_i = \frac{S_i}{S_{i-1}} = e^{r_i}, \qquad r_i = \log(1 + R_i), \qquad R_i = e^{r_i} - 1.$$

Telescoping gives the exact aggregation rules

$$1 + R_{0:n} = \frac{S_n}{S_0} = \prod_{i=1}^{n}(1 + R_i), \qquad r_{0:n} = \log \frac{S_n}{S_0} = \sum_{i=1}^{n} r_i.$$

For two periods, $R_{0:2} = R_1 + R_2 + R_1 R_2$. Adding simple returns omits the compounding term. For example, gains of 10% followed by losses of 10% produce $1.1(0.9) - 1 = -1\%$, whereas their sum is zero.

(b) If the observed returns have sample mean $\bar{r}$ and positive sample standard deviation s_r, use $z_i = (r_i - \bar{r})/s_r$. Compare the sorted z_i with $\Phi^{-1}((i - 1/2)/n)$ and draw $y = x$. A density histogram has total area one. A count histogram has a different vertical scale; a density overlay must either accompany density normalization or be multiplied by the sample size and bin width when equal-width count bins are used. An unnormalized count curve cannot be compared directly with a density histogram.

(c) Permutation preserves the multiset of returns, the empirical marginal distribution, sample mean, variance, skewness, kurtosis, and standardized tail counts. It can change adjacency, clusters, autocorrelations, and their squared-return counterparts. Small return autocorrelations do not exclude dependence of squares. For an explicit stationary example, let Z_t be independent $N(0, 1)$ and put $r_t = Z_t Z_{t-1}$. Then $\mathbb{E}r_t = 0$, $\mathrm{Var}(r_t) = 1$, and $\mathrm{Cov}(r_t, r_{t-1}) = 0$, but

$$\mathrm{Cov}(r_t^2, r_{t-1}^2) = \mathbb{E}[Z_t^2 Z_{t-1}^4 Z_{t-2}^2] - 1 = 3 - 1 = 2 > 0.$$

A single permutation is a diagnostic comparison, not a significance test.

1.5 T5. Why diffusion noise scales as the square root of time

(a) With $n = T/h$ independent standard normal variables, the accumulated noise $U_n = \sum_{i=1}^{n} \sigma h^{\alpha} Y_i$ satisfies

$$\mathbb{E}U_n = 0, \qquad \mathrm{Var}(U_n) = n\sigma^2 h^{2\alpha} = \sigma^2 T h^{2\alpha-1}.$$

As $h \downarrow 0$, a finite nonzero variance requires $2\alpha - 1 = 0$, hence $\alpha = 1/2$. For $\alpha > 1/2$ the variance vanishes; for $\alpha < 1/2$ it diverges.

(b) The following is a local Taylor heuristic on paths whose multipliers remain positive, not an unconditional finite-h real-log moment calculation. For the Euler multiplier, write $\epsilon_i = \mu h + \sigma\sqrt{h} Y_i$. In

$$\sum_i \log(1+\epsilon_i) \simeq \sum_i \epsilon_i - \frac{1}{2}\sum_i \epsilon_i^2,$$

the noise sum has variance $\sigma^2 T$, and the leading quadratic correction is $\sigma^2 T/2$. Thus the leading log mean and variance are $(\mu - \sigma^2/2)T$ and $\sigma^2 T$. Terms involving $\mu^2 h^2$ vanish after summation in this heuristic.

For the incorrect multiplier, $\widetilde{\epsilon}_i = h(\mu + \sigma Y_i)$. Its noise sum has variance $\sigma^2 T h$, and the mean quadratic contribution is $(\mu^2 + \sigma^2)Th/2$. Its formal leading log mean and variance are consequently

$$\mu T - \tfrac{1}{2}(\mu^2 + \sigma^2)Th + o(h), \qquad \sigma^2 Th + o(h).$$

The stochastic contribution disappears as the grid is refined, so the second scheme loses the diffusion term rather than approximating GBM.

(c) At every fixed $h > 0$, normal noise has an unbounded lower tail. For $\sigma > 0$, the probabilities of a nonpositive one-step multiplier are respectively

$$p_E = \Phi\Big(-\frac{1+\mu h}{\sigma\sqrt{h}}\Big) > 0, \qquad p_W = \Phi\Big(-\frac{1+\mu h}{\sigma h}\Big) > 0.$$

Therefore real logarithms can fail; conditioning on positive multipliers would change the distribution and its moments. A rigorous argument for the wrong scheme uses products directly. Put $a_h = 1+\mu h$ and $A_i = a_h + \sigma h Y_i$. Independence yields

$$\mathbb{E}\widetilde{S}_n = S_0 a_h^n, \qquad \mathbb{E}\widetilde{S}_n^2 = S_0^2 (a_h^2 + \sigma^2 h^2)^n.$$

Since $n = T/h$, these converge to $S_0 e^{\mu T}$ and $S_0^2 e^{2\mu T}$: for the second limit, its base is $1 + 2\mu h + (\mu^2 + \sigma^2)h^2$. If $c = S_0 e^{\mu T}$, then

$$\mathbb{E}[(\widetilde{S}_n - c)^2] = \mathbb{E}\widetilde{S}_n^2 - 2c\mathbb{E}\widetilde{S}_n + c^2 \longrightarrow 0.$$

Thus the terminal value converges to c in mean square and hence in probability, without any logarithms or positivity conditioning.

1.6 T6. GBM moments and a terminal prediction interval

(a) Set $m = \mu - \sigma^2/2$. The model gives $\log(S_T/S_0) = mT + \sigma\sqrt{T}Z$ for $Z \sim N(0,1)$. Using the stated normal exponential moment,

$$\mathbb{E}S_T^k = S_0^k \exp\Big(k\mu T + \tfrac{1}{2}k(k-1)\sigma^2 T\Big).$$

Taking $k = 1, 2$ and subtracting the squared mean gives

$$\mathbb{E}S_T = S_0 e^{\mu T}, \qquad \mathrm{Var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1), \qquad \mathrm{median}(S_T) = S_0 e^{mT}.$$

The median follows because Z has median zero and the exponential is increasing. For $S_0 = 100$, $\mu = 0.08$, $\sigma = 0.30$, and $T = 2$, $m = 0.035$ and

$$\mathbb{E}S_T = 117.3510871, \quad \mathrm{Var}(S_T) = 2715.9350636, \quad \mathrm{median}(S_T) = 107.2508181.$$

The price standard deviation is 52.1146339 price units; the variance has squared price units.

(b) A central 95% model prediction interval is

$$\left[S_0 e^{mT - 1.96\sigma\sqrt{T}},\ S_0 e^{mT + 1.96\sigma\sqrt{T}}\right] = [46.6938569,\ 246.3437112].$$

It is obtained by transforming the standard-normal quantiles monotonically. The use of 1.96 makes the coverage approximately 95%.

(c) This interval describes the randomness of one future price under fixed known model parameters. It is not an interval estimating a population mean or an unknown parameter. For M independent Monte Carlo terminal draws, the estimated mean instead has standard error $\mathrm{sd}(S_T)/\sqrt{M}$, estimated by $s_S/\sqrt{M}$. Its approximate 95% mean confidence interval is $\bar{S} \pm 1.96 s_S/\sqrt{M}$. Increasing M reduces this simulation uncertainty; it does not narrow the fixed-model prediction interval or eliminate parameter/model uncertainty.

1.7 T7. Long-run growth versus expectation

(a) Divide the exact logarithmic solution by t:

$$\frac{1}{t}\log\frac{S_t}{S_0} = \mu - \frac{\sigma^2}{2} + \sigma\frac{W_t}{t} \longrightarrow g := \mu - \frac{\sigma^2}{2} \quad \text{almost surely.}$$

If $g < 0$, eventually the logarithm is bounded above by a negative constant times t, so $S_t \to 0$ almost surely. If $g > 0$, it grows at least as a positive constant times t, so $S_t \to \infty$ almost surely.

(b) Stock A has $g_A = 0.10 - 0.50^2/2 = -0.025$ and tends to zero almost surely. Stock B has $g_B = 0.06 - 0.30^2/2 = 0.015$ and tends to infinity almost surely. A has the higher arithmetic drift but lower typical compound growth. Random percentage losses and gains compound asymmetrically; the quadratic volatility correction reduces long-run log growth. These are conclusions of the perpetual constant-parameter teaching model.

(c) For A, $\mathbb{E}S_t = S_0 e^{0.10t}$ still grows. For any fixed finite t, rare very large prices can contribute strongly to the mean while most paths eventually become small. Almost-sure convergence alone does not justify interchanging expectation and an infinite-time limit; the relevant family is not uniformly integrable in this case.

At the critical value $\mu = \sigma^2/2$ with $\sigma > 0$, $S_t = S_0 e^{\sigma W_t}$ and the asymptotic log growth is zero. A zero rate says only that the logarithm grows sublinearly in t; it does not say the logarithm converges. Indeed, for every fixed $K > 0$,

$$\mathbb{P}(S_t \le K) = \Phi\left(\frac{\log(K/S_0)}{\sigma\sqrt{t}}\right) \longrightarrow \frac{1}{2}.$$

This rules out convergence in probability, and therefore almost-sure convergence, to any finite deterministic constant: for a proposed constant $c \ge 0$, choose $K > c$, for which such convergence would require the probability to tend to one. Brownian randomness remains at the critical drift even though its contribution divided by t vanishes.

1.8 T8. Calibrating a constant-parameter return model

(a) The model implies $\mathbb{E}r_i = mh$ and $\mathrm{Var}(r_i) = \sigma^2 h$, where $m = \mu - \sigma^2/2$. Matching these moments to $\bar{r}$ and the sample variance with divisor $n-1$ gives, for $n \geq 2$,

$$\widehat{m} = \frac{\bar{r}}{h}, \qquad \widehat{\sigma^2} = \frac{s_r^2}{h}, \qquad \widehat{\sigma} = \frac{s_r}{\sqrt{h}}, \qquad \widehat{\mu} = \frac{\bar{r}}{h} + \frac{s_r^2}{2h}.$$

For $h = 1/252$, these become $252\bar{r}$, $252s_r^2$, $\sqrt{252}s_r$, and $252\bar{r} + 126s_r^2$, respectively. Returns are dimensionless; m and μ are rates per year, σ is per square root of a year, and σ^2 is per year. A daily mean should therefore be multiplied by 252, whereas a daily standard deviation is multiplied by $\sqrt{252}$.

These are moment estimators using an unbiased sample variance. In the i.i.d.-normal model, the MLE of m is still $\bar{r}/h$, but the variance MLE is

$$\widehat{\sigma^2}_{\mathrm{ML}} = \frac{1}{nh}\sum_{i=1}^{n}(r_i - \bar{r})^2 = \frac{n-1}{n}\widehat{\sigma^2}.$$

The volatility and arithmetic-drift MLEs use that variance MLE, so the displayed moment estimators should not all be labeled MLEs.

(b) Independence gives

$$\mathrm{Var}(\widehat{m}) = \frac{1}{h^2}\mathrm{Var}(\bar{r}) = \frac{\sigma^2}{nh}, \qquad \mathrm{SE}(\widehat{m}) = \frac{\sigma}{\sqrt{nh}}.$$

With $H = nh$ observed years, the plug-in standard error is

$$\widehat{\mathrm{SE}}(\widehat{m}) = \frac{\widehat{\sigma}}{\sqrt{H}} = \frac{s_r}{h\sqrt{n}}.$$

Increasing observation frequency at fixed H does not reduce the model's drift standard error $\sigma/\sqrt{H}$. Each shorter increment supplies a smaller drift signal; more observations are not equivalent to observing more years. This result assumes exact nonoverlapping observations from the stated model; market microstructure can make very high-frequency observations depart from it.

(c) First, parameters or the data-generating mechanism can change after the calibration period. Second, a marginal Q–Q plot does not check serial independence, constant conditional variance, or the absence of jumps and other dynamics. Finite-sample estimates, the selected adjustment field, and data errors introduce further uncertainty. A fitted plot is a diagnostic for a historical sample, not proof of the law of future returns.

2 Part II: Algorithms

2.1 A1. Five-year real-data return analysis

(a) Input and audit. MSFT (Nasdaq), USD; Yahoo Finance public historical/chart interface; retrieved on 21 October 2026. The frozen window is 14 October 2021–14 October 2026, inclusive. The selected adjusted-close field includes the provider's split and dividend adjustments as of retrieval. It describes an adjusted return series, not a raw execution price. The input has 1255 observations and 1254 adjacent log returns. There are zero duplicate dates, missing values, nonpositive prices, missing expected trading sessions, or unexpected observations. The longest calendar gap is four days. Weekends, exchange holidays and the 9 January 2025 Carter closure were audited; no nontrading dates were forward-filled.

The analyzed CSV SHA-256 is

```
ce807ba43ff39e775faebda202ad5d8476569efa74973284a93216fbae8eff9f
```

The exact raw-response hash, retrieval timestamp, request URL, currency and field definition are retained in `data/provenance.json`. Raw data remain in the private local data directory; redistribution rights were not established, so no public-data redistribution right is asserted. The downloader and public historical interface provide a course-access route. Any later provider revision must be identified by a changed hash, rather than presented as the same input.

(b) Implemented statistics. Let n be the number of returns, $y_i = r_i - \bar{r}$, and $m_k = n^{-1}\sum y_i^k$. The variance uses $s^2 = \sum y_i^2/(n-1)$; skewness is $m_3/m_2^{3/2}$ and excess kurtosis is $m_4/m_2^2 - 3$. The last two are uncorrected central-moment ratios, not bias-adjusted estimators.

Quantity	Observed value
Number of returns	1254
Daily mean	0.000484860
Daily variance	0.000312370
Daily standard deviation	0.017674004
Skewness	0.356888
Excess kurtosis	6.165976

The histogram uses density normalization. The fitted normal uses $\bar{r}, s_r$; the Q–Q plot uses sorted $(r_i - \bar{r})/s_r$ and theoretical quantiles $\Phi^{-1}((i-1/2)/n)$ with the reference $y = x$.

(c) Tail comparison. The counts use the strict inequality $|r_i - \bar{r}| > ks_r$; fitted-normal expected counts are $2n[1 - \Phi(k)]$.

Threshold k	Observed	Normal expected
3	13	3.3855443
4	6	0.0794315
5	3	0.0007189

The fitted normal is a rough central reference. The Q–Q tails depart from its line and the observed tail counts exceed its expectations; this sample therefore does not support an exact fitted i.i.d.-normal description. These are fitted in-sample comparisons, not exact hypothesis tests: estimated standardization, dependence, parameter changes, dividend adjustments and a few extreme observations matter. No particular alternative tail law or market inefficiency follows.

(d) Temporal diagnostics. The code implements the exact global-mean/global-denominator ACF given in the handout. The same single seeded permutation is used for both return and

squared-return comparisons. Returns have small ACFs at lags 1–20 (largest absolute value about 0.0524); some squared-return ACFs are positive, reaching about 0.0709. These observations neither prove independence nor establish a volatility model. One permuted squared-return ACF reaches about 0.1287, illustrating why one permutation is not a significance test. Permutation preserves the complete empirical marginal distribution, moments, tail counts and compounded final return; it changes ordering, intermediate paths and temporal dependence.

(e) Calibration with $h = 1/252$.

Estimate	Value
$\widehat{m}$ (year^{-1})	0.122185
$\widehat{\sigma}$ (year$^{-1/2}$)	0.280566
$\widehat{\sigma}^2$ (year^{-1})	0.078717
$\widehat{\mu}$ (year^{-1})	0.161543
Plug-in SE($\widehat{m}$) (year^{-1})	0.125773
$H = nh$ (years)	4.976190

The log-drift SE is about as large as its estimate. The physical drift is imprecisely estimated, and these parameters are descriptive constant-model estimates, not forecasts guaranteed by the data. Dividend-adjusted prices and annualization conventions must be kept consistent.

Checks include strict positive/finite prices and duplicate/date validation; exchange-calendar coverage and no forward filling; telescoping log-return/compounding identities; hand-calculated ACF and moment divisors; and preservation of marginal statistics under permutation. Commands and actual results are preserved alongside the code.

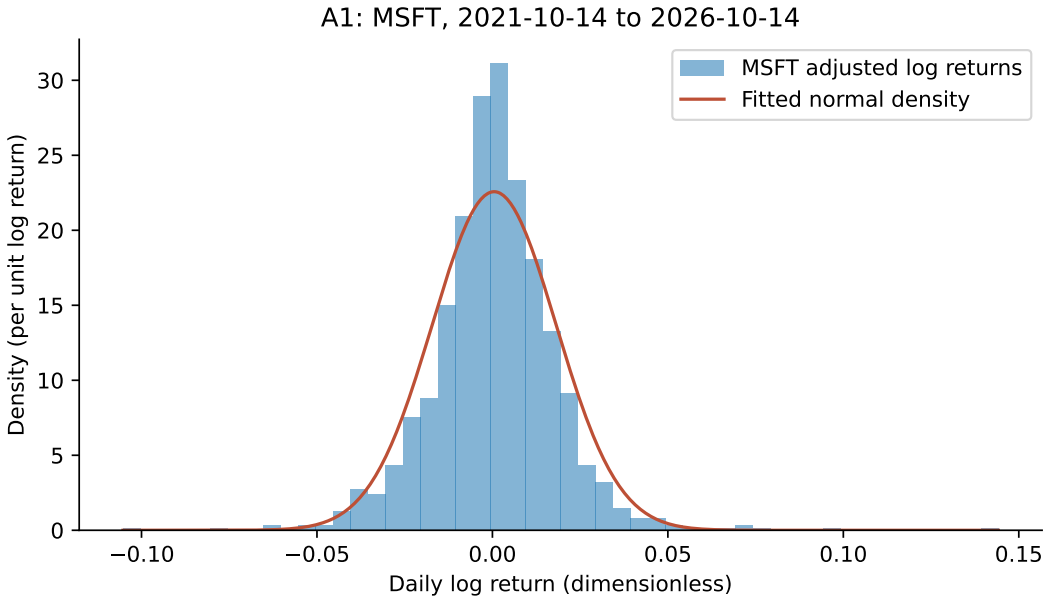


Figure 1: Density histogram and fitted normal for daily adjusted MSFT log returns.

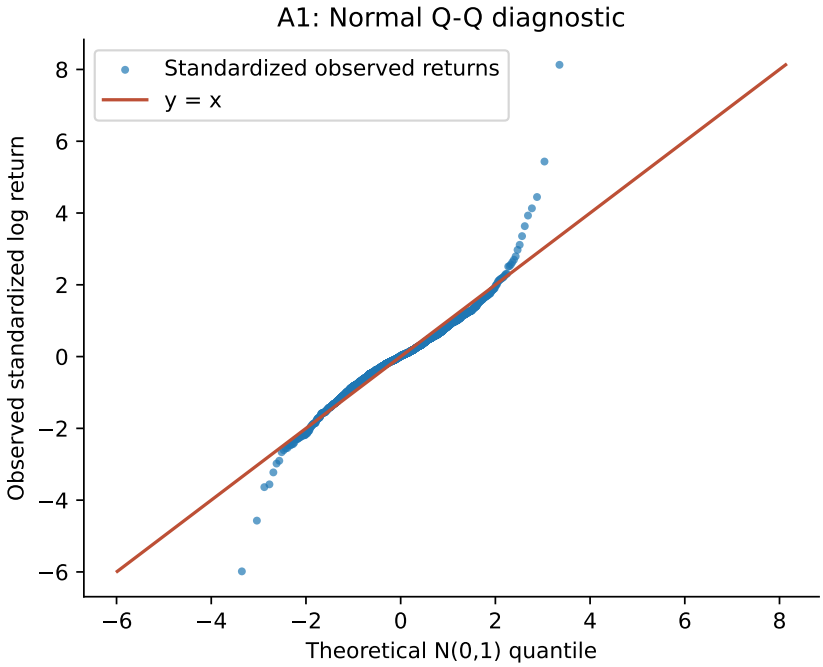


Figure 2: Standardized-return normal Q–Q diagnostic. Extremes depart from the fitted Gaussian benchmark.

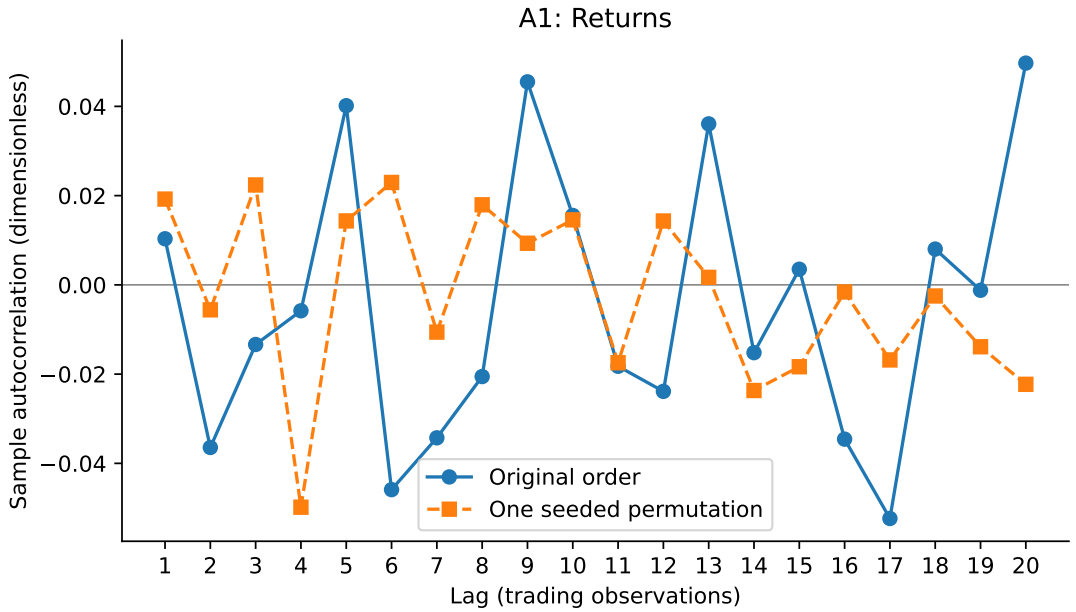


Figure 3: Return ACF: original order and one seeded permutation.

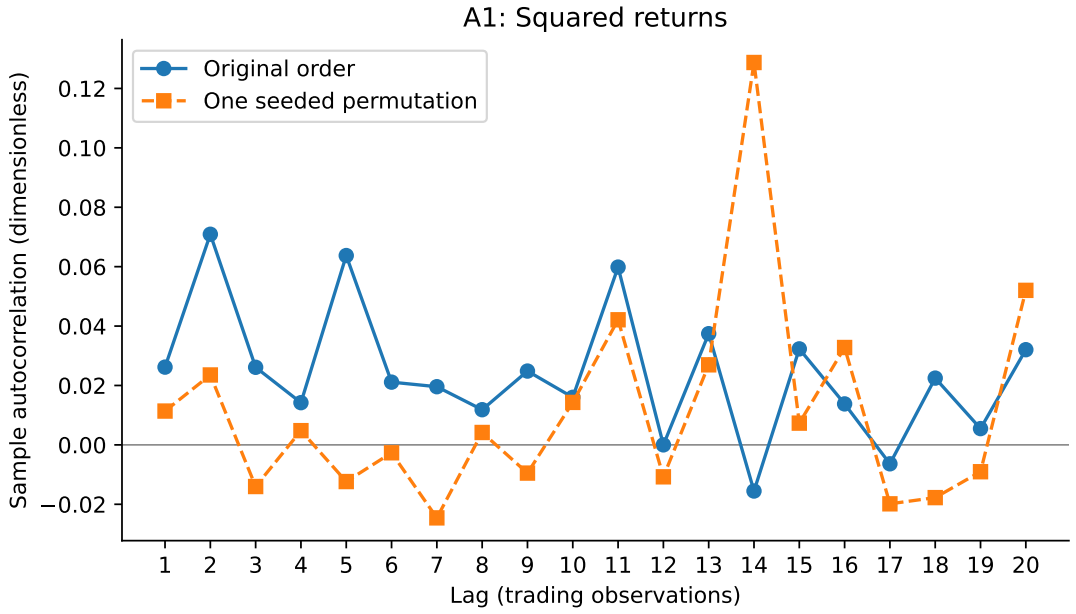


Figure 4: Squared-return ACF for the same two orders; neither curve is a significance threshold.

2.2 A2. Lognormal densities and exact terminal simulation

(a) Density and normalization. For $T > 0$, $\sigma > 0$,

$$f_T(s) = \frac{1}{s\sigma\sqrt{2\pi T}} \exp\left\{ -\frac{[\log(s/S_0) - (\mu - \sigma^2/2)T]^2}{2\sigma^2 T} \right\}, \quad s > 0,$$

and $f_T(s) = 0$ for $s \le 0$. This follows by transforming the normal log price with Jacobian $1/s$. Elementary quadrature on $(0, \infty)$ gives integrals 1.0000000000000007 and 1.0000000000000002 for $S_0 = T = 1$, $\sigma = 0.4$, $\mu = 0.05, 0.20$. Increasing μ shifts the log-price location upward; under common Z, it multiplies S_T by $e^{\Delta\mu T}$ while preserving log variance.

(b) Exact simulation. Draw $Z_j \sim N(0,1)$ independently and set

$$S_T^{(j)} = S_0 \exp[(\mu - \sigma^2/2)T + \sigma\sqrt{T}Z_j].$$

No Euler stepping or library lognormal sampler is used. At the T6 parameters, $M = 100{,}000$ gives

Quantity	Simulation	Analytical
Terminal mean	117.188510	117.351087
Terminal variance	2707.881389	2715.935064
Mean MCSE	0.164556	0.164801

The approximate mean confidence interval is $\bar{S} \pm 1.96 s_S/\sqrt{M}$:

$$[116.865980, 117.511041].$$

The mean discrepancy is about -0.988 estimated MCSEs. Sampling variation is expected; a single nominal 95% interval miss alone would not identify a program defect.

(c) Fixed prediction-interval coverage. T6's endpoints remain $[46.693857, 246.343711]$. The observed count inside is 95060 out of 100,000, or 0.950600. With rounded $z = 1.96$, the exact model probability is $p_\star = 2\Phi(1.96) - 1 = 0.95000421$, and

$$\mathrm{SD}(\widehat{p}) = \sqrt{p_\star(1 - p_\star)/M} = 0.000689175.$$

The observed discrepancy is about 0.8645 MCSEs. This binomial error scale measures the coverage fraction; it is not uncertainty in the terminal mean.

For $T = 0$, the terminal price is S_0; for $\sigma = 0$, it is $S_0 e^{\mu T}$. Both are point masses and have no ordinary lognormal density. The sampler and moment routines support them, while the ordinary-density routine raises an explicit error. Nonfinite inputs, nonpositive S_0, negative volatility/time and invalid counts are checked.

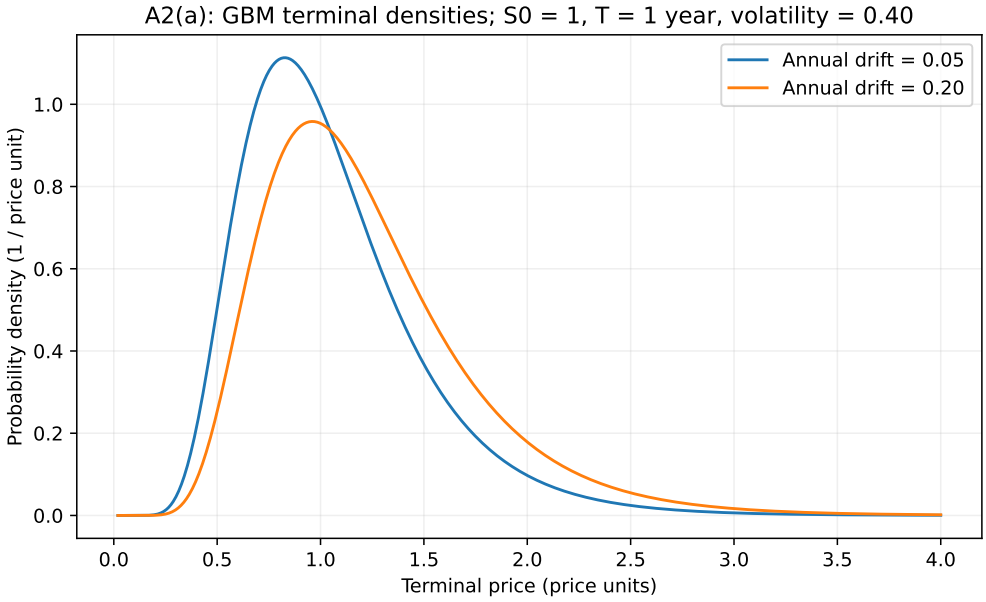


Figure 5: Analytical densities with common volatility and two physical drifts.

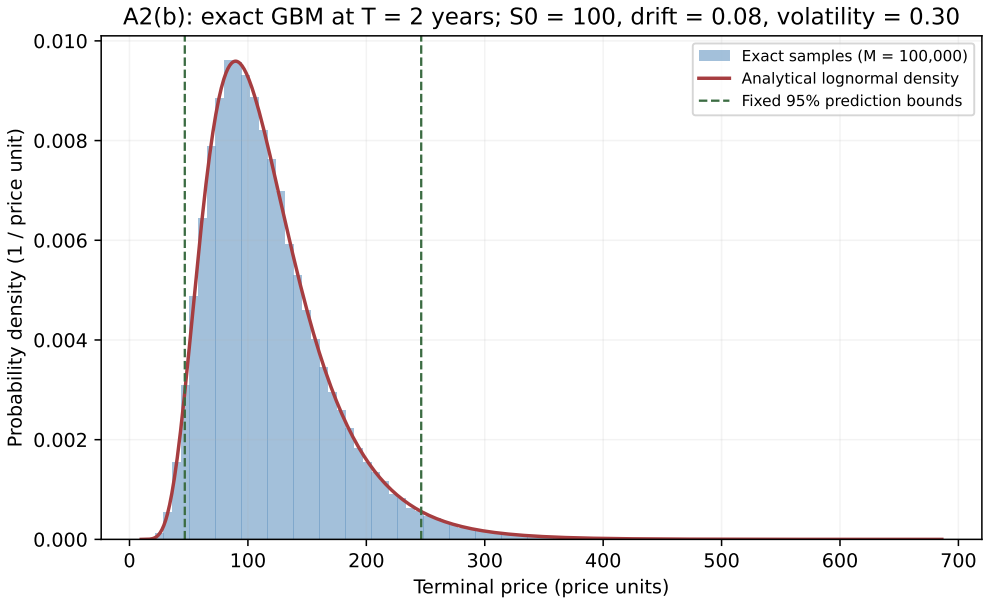


Figure 6: 100,000 exact terminal prices, density-normalized histogram and analytical density.

2.3 A3. Coupled discretizations and vanishing noise

(a) Coupling and updates. Each path uses one 256-increment Brownian grid. For $n \in \{16, 32, 64, 128, 256\}$, consecutive fine increments are summed in blocks of $256/n$; consequently every resolution has the same terminal Brownian increment. With $h = 1/n$ and coarse ΔW_i, the implementations are

$$\begin{aligned} G_T &= S_0 e^{(\mu-\sigma^2/2)T+\sigma\sum_i \Delta W_i}, \\ E_{i+1} &= E_i(1 + \mu h + \sigma\Delta W_i), \\ V_{i+1} &= V_i(1 + \mu h + \sigma\sqrt{h}\,\Delta W_i). \end{aligned}$$

Since $\Delta W_i = \sqrt{h}Y_i$, the final line is exactly the specified wrong $\sigma h Y_i$ scheme. All 20,000 paths are retained. Nonpositive multipliers are counted before updating, with no clipping or removal.

(b) Independent analytical benchmarks. Let $a = 1 + \mu h$ and let c be the Brownian coefficient (σ for Euler, $\sigma\sqrt{h}$ for the wrong scheme). For $X_n = S_0 \prod_i (a + c\Delta W_i)$,

$$\begin{aligned} \mathbb{E}X_n &= S_0 a^n, & \mathbb{E}X_n^2 &= S_0^2(a^2 + c^2 h)^n, \\ \mathrm{Var}(X_n) &= S_0^2[(a^2 + c^2 h)^n - a^{2n}], \\ \mathbb{E}[X_n G_T] &= S_0^2 e^{\mu T}(a + c\sigma h)^n. \end{aligned}$$

The last identity follows by differentiating the Gaussian exponential moment. It independently checks coupling. Substituting the second and cross moments into $\mathbb{E}[(X_n - G_T)^2] = \mathbb{E}X_n^2 + \mathbb{E}G_T^2 - 2\mathbb{E}[X_n G_T]$ yields an exact finite-n RMSE benchmark. The wrong scheme's distance from $d = S_0 e^{\mu T}$ is $\mathrm{Var}(V_n) + (\mathbb{E}V_n - d)^2$ before taking the square root.

Here are the actual terminal moments; means and errors are in price units, variances in squared price units.

Method	n	Mean	Sample variance	Product variance
euler	16	108.1115	1086.6859	1090.0748
wrong	16	108.2544	65.4672	65.4996
euler	32	108.1063	1091.5136	1097.5659
wrong	32	108.2800	32.8573	32.8786
euler	64	108.1074	1095.2981	1101.3445
wrong	64	108.2964	16.4642	16.4710
euler	128	108.1051	1096.2403	1103.2422
wrong	128	108.3069	8.2411	8.2434
euler	256	108.1093	1097.4422	1104.1931
wrong	256	108.3139	4.1230	4.1237

The common exact reference has sample mean 108.112308 and variance 1098.854857; the GBM population values are 108.328707 and 1105.145457. Monte Carlo variability explains their differences. The detailed CSV also reports mean, variance, cross-moment and squared-error uncertainty scales.

n	Euler vs. exact	Wrong vs. exact	Wrong vs. deterministic
16	1.791140	25.196830	8.091317
32	1.272735	27.521571	5.732196
64	0.896502	29.169397	4.057641
128	0.636632	30.335608	2.870737
256	0.448867	31.160114	2.030526

Regressing log(RMSE) on $\log h$ gives an observed Euler slope of 0.499243. This run is consistent with order 1/2 strong convergence; an observed rate is not a proof. The wrong variance decreases toward zero and its deterministic-target RMSE falls, while its coupled GBM error increases toward $\sqrt{\mathrm{Var}(G_T)} \approx 33.244$.

(c) One-step stress test. At $S_0 = 100$, $\mu = .08$, $\sigma = T = h = 1$,

$$\mathbb{P}(1 + .08 + Y \leq 0) = \Phi(-1.08) = 0.1400710901.$$

There are 14,178 nonpositive outcomes among 100,000 draws, fraction 0.141780. The model binomial MCSE is 0.0010975025; the discrepancy is about 1.5571 MCSEs. Clipping would replace the simulated law and corrupt both this probability diagnostic and moment/error calculations.

(d) Cost and memory. Batches have at most $B = 1000$ paths. Runtime is $O(M[n_{\max} + \sum n])$, or $O(Mn_{\max})$ for these geometric resolutions. Peak working-array storage is $O(Bn_{\max})$: the finest-grid array uses 2,048,000 bytes and the conservative dominant-array estimate is 4,192,000 bytes. Retained terminal outputs use an additional $O(MK)$ space, 1,760,000 bytes for $K = 5$ resolutions; no full path history is saved. These are array sizes, not measured total process memory.

Using an independent exact path would not measure pathwise approximation error. Even if Euler converged to one exact realization, an independent exact reference would leave limiting MSE $2\,\mathrm{Var}(G_T)$ rather than zero. The common Brownian grid removes this unrelated physical variation.

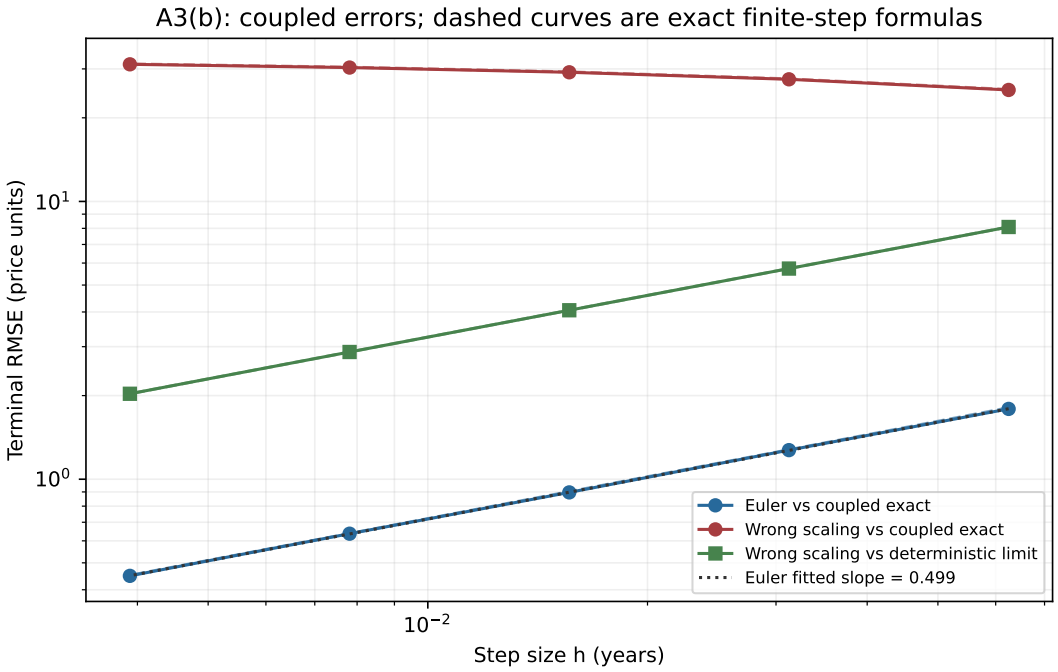


Figure 7: Log–log coupled errors: Euler approaches GBM, while the wrong scheme approaches the deterministic target.

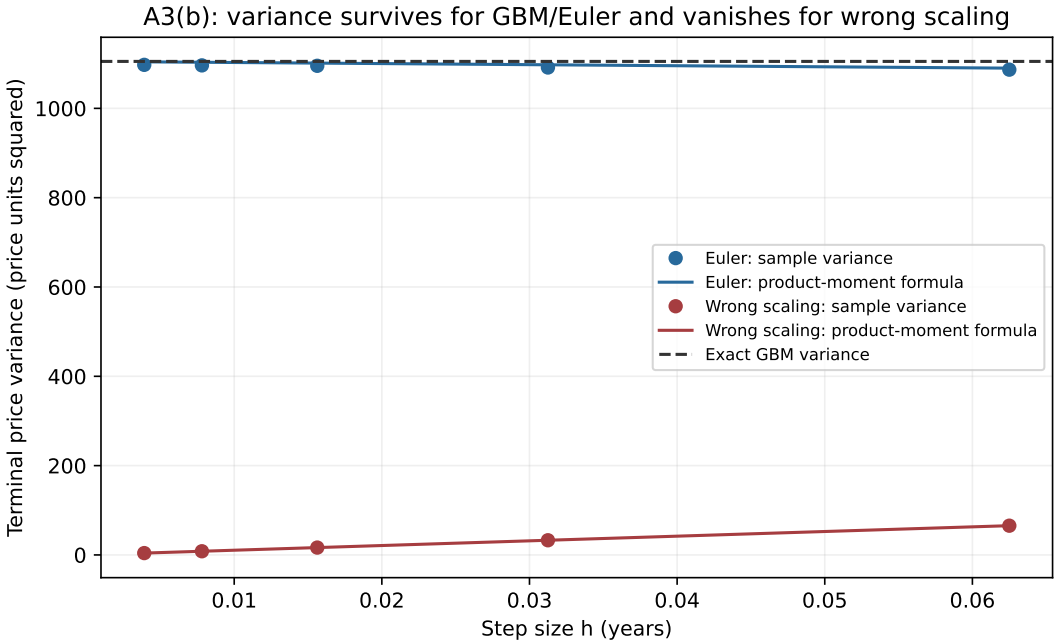


Figure 8: Sample and exact finite-grid product variances; the wrong noise disappears as the grid is refined.

3 Part III: AI interaction

3.1 I1. Two-session audit

Two independently prompted DeepSeek sessions, Model A and Model B, received the same four claims and mathematical conventions on 21 October 2026. The interface did not expose distinct backend build identifiers, so this is not a claim of two vendor-distinct models; it is a two-session audit within one provider. Earlier draft labels naming OpenAI models were incorrect and have been corrected. Neither session browsed or executed code while answering; the orchestrator ran the verification independently. The prompts were issued by the orchestrating assistant on the student's behalf, not typed personally by the student. Full initial prompts, actual replies and two substantive follow-ups per session are preserved separately.

Claim	A	B	Independent conclusion
Efficiency implies i.i.d. Gaussian	Invalid	Invalid	Mean restrictions do not fix a law
$\mu > 0$ implies a.s. growth	Invalid	Invalid	Need $\mu > \sigma^2/2$
More paths narrow a fixed PI	Invalid	Invalid	PI independent of replication count
Smooth $\sigma h Y_i$ verifies GBM	Invalid	Invalid	Noise variance $\sigma^2 T h \to 0$

Hand argument 1. Dividing $\log(S_t/S_0)$ by t and using $W_t/t \to 0$ gives $\mu - \sigma^2/2$. With $\mu = .10, \sigma = .50$, this is $-.025$: the price tends to zero almost surely although its expectation grows.

Hand argument 2. For the wrong scheme, independence gives the exact first and second product moments in T5. They converge to d and d^2, so $\mathbb{E}[(V_n - d)^2] \to 0$. This proof uses no invalid logarithm of a potentially negative multiplier.

Numerical check. A3 uses 20,000 coupled paths. Refining from 16 to 256 steps reduces Euler RMSE from 1.791140 to 0.448867, while the wrong scheme's variance drops from 65.467154 to 4.123023 and its GBM RMSE increases from 25.196830 to 31.160114. The independent finite-grid product/cross-moment checks agree within their Monte Carlo scales. This directly distinguishes visual smoothness from convergence to GBM.

Response-specific follow-ups. Model A initially illustrated mean-unpredictability with an unbounded product-normal sequence. A follow-up identified that it can be below -1 if called a positive-stock simple return. It supplied a bounded discrete martingale-difference construction with strictly positive gross returns and positively dependent squared returns. This is a qualified counterexample, not a claim that economic efficiency was proved. Its second follow-up derived the coupled cross-moment benchmark used in A3. Model B supplied a complete product-moment proof after its first response had relied on summed-noise intuition; its second follow-up clarified the Jensen/risk-benchmark qualification of a symmetric log-return example and computed MC uncertainty scaling.

Comparison (206 words). Both models rejected the four claims, so disagreement was not needed to make the audit useful. Model A gave a detailed distinction between conditional-mean restrictions and distributional assumptions, including an explicit dependent counterexample. Its initial product-normal example, however, could not be used unchanged as the simple return of a positive-price stock. A targeted follow-up produced a bounded construction and preserved the original limitation. Model B was more concise. Its initial argument about vanishing summed noise needed a product-moment justification for the multiplicative scheme; the follow-up supplied that proof. Its symmetric log-return example also required an economic benchmark because a zero expected log return does not mean a zero expected simple return.

Cochrane's discussion of Fama, actually consulted through Chicago Booth Review, emphasizes the relation between available information, trading rules, risk, and empirical evaluation. That

perspective supports treating efficiency as an economic claim requiring a benchmark, rather than equating it with Gaussian noise. Neither model response establishes empirical efficiency. The useful difference was explanatory depth and which assumptions became explicit after questioning. Mathematical derivations and A3's independently executed coupled experiment provided the verification; fluent agreement alone did not. The record therefore reports qualified examples and completed checks without inventing an incorrect classification to create a model error.

Non-AI source: John H. Cochrane, "Eugene Fama, Efficient Markets, and the Nobel Prize," Chicago Booth Review, 20 May 2014, https://www.chicagobooth.edu/review/eugene-fama-efficient-markets-and-the-nobel-prize. The Fama 1970 bibliographic record was checked; its full text was not claimed to have been read.

3.2 I2. Repair of the instructor terminal-price routine

The unchanged original was supplied in `code.zip` and preserved byte-for-byte, SHA-256

```
7f9bf4b6a5621cfdc427ecf0cf56621ff37c9a5662028570793a29aea8adf174
```

The original is deliberately flawed instructor-authored teaching code; it is not attributed to an AI. The student reported having no earlier independent/pre-AI diagnostic record. The saved initial diagnosis and the 11-check design are explicitly an assistant baseline written before the separate repair-model request; they cannot satisfy a personal pre-AI chronology retrospectively. Rubric item (a) is therefore only partly met: the diagnosis and checks exist and precede the repair request, but they were not authored personally by the student. This gap is disclosed rather than backdated.

The original exponent uses $\mu T + \sigma T Z$, omitting $-\sigma^2 T/2$ and replacing $\sqrt{T}$ by T. It computes SE as s/M instead of $s/\sqrt{M}$, and labels a narrow interval for the simulated mean as a terminal prediction interval. It also lacks model/count validation. Correct code uses the exact A2 law, sample variance with divisor $M-1$, an explicitly named mean confidence interval, and analytical fixed prediction endpoints. Deterministic limits return exact zero variance and SE.

Required category	Executed checks (fixed script)	Count
Analytical benchmark	mean within 5 model MCSE; variance within 5%; same-draw drift/time; divisor $M-1$	4
Uncertainty/interval	SE $= s/\sqrt{M}$; fixed population PI; separate mean CI; quadrupling M halves SE	4
Invalid/limiting input	$T=0$ point mass; $\sigma=0$ point mass; invalid model/count rejected	3

The same 11 checks in `code/debug_checks.py` were run unchanged against all three versions, so the before/after comparison does not use tests written after seeing a patch.

The unchanged original passed 1 of the 11 initial checks. At the default parameters it returned mean 140.073177, variance 8452.529174, SE 0.0009193764, and a purported prediction interval [140.071375,140.074979], instead of the T6 distribution and uncertainty scales. The final repair passes 11 of 11 checks. Its default mean is 117.100664, sample variance 2701.350850, and mean MCSE 0.164357867. Its population prediction interval is [46.693857,246.343711], while its mean confidence interval is [116.778523,117.422806]. The seed differs from the A2 child stream; equality of sample outcomes is not expected. A separate same-draw check agrees with the A2 sampler to numerical precision.

Three substantive repair exchanges, including the actual candidate test output and a more demanding follow-up, are preserved in `ai_logs/I2_model_B.md`. The original, proposed patch and final repair remain separate. A passing candidate is reported honestly; no artificial model failure is

invented. Accepted/revised decisions and all executed checks are recorded in the repair log. The first proposal passed 8/11 direct checks: its numerical formulas were correct, but three checks failed because it renamed the original `prediction_interval` key. Restoring that key as an alias of the explicitly named fixed-population interval resolved the compatibility regression. After 11/11 passed, 44 more demanding checks also passed: 36 same-draw comparisons across three drifts, three volatilities and four horizons, one prediction-interval/count invariance check, and seven invalid-input cases.

Accepted, revised and rejected changes. Accepted: the $-\sigma^2 T/2$ log-drift correction, $\sigma\sqrt{T}$ noise scaling, divisor $M-1$, SE $= s/\sqrt{M}$, separation of the mean confidence interval from the fixed population prediction interval, input validation, and exact deterministic limits. Revised: the proposed removal of the `prediction_interval` key was rejected as an API break and replaced by a backward-compatible alias with identical endpoints in every branch. Rejected: treating the three `KeyError` failures as numerical errors (they were not); changing formulas after the 11/11 pass with no failing evidence; and inventing a further failure to make the interaction look more eventful. The recorded 1/11, 8/11 and 11/11 outcomes are the actual executed results.

These checks establish tested model behavior, not empirical validity of GBM or numerical representability for every extreme input.

4 Part IV: Reading notes and cross-book comparison

R1 reading notes, page 1 of 4: Tsay—returns, aggregation, and moments

Source and disclosure. Ruey S. Tsay, *Analysis of Financial Time Series*, third edition, Wiley, 2010, ISBN 9780470414354. The publisher's public chapter excerpt was consulted: https://catalogimages.wiley.com/images/db/pdf/9780470414354.excerpt.pdf. Printed pp. 2–23 were used: §1.1 pp. 2–6, §1.2 pp. 7–22, and §1.3 pp. 22–23. The notes are AI-assisted; they record a source-based close reading, not a claim that the student personally completed the reading.

Concepts and assumptions. Tsay begins with prices P_t, simple returns R_t, and log returns r_t, then treats sample moments, marginal and conditional distributions, and the distinction between an empirical distribution and a model. The algebra on pp. 3–6 requires positive prices but not normality, independence, or constant volatility. The distributional discussion on pp. 7–23 is diagnostic: it asks what a return sample looks like and what time dependence it exhibits, rather than asserting a data-generating law.

Reconstruction: telescoping and the missing cross term

For positive prices with no dividend inside the interval,

$$1 + R_t = \frac{P_t}{P_{t-1}}, \qquad r_t = \log \frac{P_t}{P_{t-1}}.$$

Intermediate prices cancel in a product, so for k periods

$$\prod_{j=1}^{k}(1 + R_j) = \frac{P_k}{P_0}, \qquad \sum_{j=1}^{k} r_j = \log \frac{P_k}{P_0}.$$

For two periods, $R_{0:2} = (1 + R_1)(1 + R_2) - 1 = R_1 + R_2 + R_1 R_2$. Adding simple returns therefore omits the cross term; adding log returns is an exact temporal identity. This is the T4(a) point and the reason A1 uses log returns while reporting compounded simple returns separately.

Checked example: Example 1.1, p. 6

Recomputing the printed percentage log returns,

$$100(e^{0.0446} - 1) = 4.56095324\%, \qquad 4.46 - 7.34 + 10.77 = 7.89\%.$$

Transforming the accumulated log return once gives

$$100(e^{0.0789} - 1) = 8.20961070\%,$$

whereas transforming each monthly return and adding gives 8.85521777%. The difference is the cross term above, not rounding error. When returns are quoted in percent, the exponential argument must first be divided by 100.

Course connection and question. This material supplies the aggregation and compounding conventions of A1 and T4; it also warns that the dividend adjustment on p. 5 changes what the price ratio measures. My unresolved question is how a provider's retrospective adjustment convention affects the economic interpretation of a historical series even though the algebra is unchanged.

R1 reading notes, page 2 of 4: Tsay—marginals, tails, and time order

Precise locations. Sample moments and divisors appear on pp. 8–9, equations (1.10)–(1.13); Example 1.2 and its calculation on pp. 10–14; the conditional-distribution factorization on p. 15, equations (1.15)–(1.16); distributional alternatives on pp. 16–18; empirical comparisons and volatility discussion on pp. 19–23. Chapter 2 was not used in these notes.

Reconstruction: a marginal distribution cannot identify dependence

Let Z_t be independent standard normals and set $X_t = Z_t Z_{t-1}$. Then

$$\mathbb{E}X_t = 0, \qquad \mathbb{E}X_t^2 = 1, \qquad \mathbb{E}(X_t X_{t-1}) = 0.$$

Nevertheless,

$$\mathbb{E}(X_t^2 X_{t-1}^2) = \mathbb{E}Z_t^2 \, \mathbb{E}Z_{t-1}^4 \, \mathbb{E}Z_{t-2}^2 = 3, \qquad \mathrm{Cov}(X_t^2, X_{t-1}^2) = 2.$$

Thus zero linear autocorrelation does not imply independence. A permutation preserves every observation, hence the empirical marginal distribution, moments, tail counts, and the compounded final return, but it changes adjacency and therefore the ACF. This is exactly what A1(d) displays when it compares a return series and one seeded permutation; one permutation is an illustration, not a significance test.

Checked examples: Example 1.2 and the variance mixture

For the reported sample size 9845 and skewness 0.06139878,

$$t = \frac{0.06139878}{\sqrt{6/9845}} = 2.487092723,$$

and the two-sided standard-normal tail probability is $p \approx 0.0128792$, agreeing with the book's displayed calculation. For the mixture illustrated on pp. 17–18 with weights $0.95, 0.05$ and component variances $1, 16$,

$$v = 0.95(1) + 0.05(16) = 1.75, \qquad \mathbb{E}X^4 = 3[0.95(1)^2 + 0.05(16)^2] = 41.25,$$

$$\frac{\mathbb{E}X^4}{v^2} - 3 = 10.46938776.$$

Independent draws from this mixture have fat marginal tails but no time dependence. Non-Gaussianity and dependence are therefore separate properties, which is why A1 reports a histogram, a Q–Q plot, and ACFs rather than treating them as substitutes.

Course connections and questions. This connects to T1 (a non-Gaussian histogram does not settle weak-form efficiency), T4(c) (permutation changes time order, not the marginal), and A1(b)–(d). My open questions are how stable these diagnostics are across subsamples and how conditional volatility should be assessed when a constant-normal calibration looks acceptable in the center but fails in the tails.

R1 reading notes, page 3 of 4: Hull—Wiener processes, Itô's lemma, and the lognormal price

Source and edition handling. The assignment specifies John C. Hull, *Options, Futures, and Other Derivatives*, eleventh U.S. edition, Pearson, copyright 2022 (published 2021), ISBN 9780136939979, Chapter 14, §§14.1–14.8 and the appendix. The publisher catalog confirms the chapter title and that the eleventh edition adds §14.8 on fractional Brownian motion. A complete eleventh-edition chapter text was *not* lawfully accessible from this environment, so no unofficial full-book copy was used. The complete text actually read was a course copy of the same Chapter 14 in the **ninth-edition printing**, pp. 302–320: §§14.1–14.7 and the appendix. All page numbers below are therefore ninth-edition pages, and equation numbers are those printed in that copy; they are not attributed to the eleventh edition. The section names are cross-checked against the publisher's eleventh-edition contents. This edition deviation is disclosed rather than concealed; §14.8 was not read. The notes are AI-assisted and are not an attestation of personal reading.

Concepts and assumptions. Hull assumes that a stock price is a Markov process (§14.1), which he relates to weak-form efficiency, and builds a continuous-time model with constant expected return μ and volatility σ (§§14.2–14.4). A Wiener process has $\Delta z = \xi\sqrt{\Delta t}$ with $\xi \sim N(0,1)$ and independent increments, so variances add while standard deviations grow like $\sqrt{T}$. A generalized Wiener process is $dx = a\,dt + b\,dz$; an Itô process lets a and b depend on x and t. The stock model is

$$\frac{dS}{S} = \mu\,dt + \sigma\,dz, \qquad \Delta S = \mu S\,\Delta t + \sigma S\,\xi\sqrt{\Delta t} \quad \text{(9th-ed. eqs. 14.6–14.8).}$$

Reconstruction: why the second derivative survives, and the lognormal law

Expand $G(S,t)$ to second order and substitute $\Delta S = \mu S\Delta t + \sigma S\xi\sqrt{\Delta t}$:

$$\Delta G = G_S\Delta S + G_t\Delta t + \tfrac{1}{2}G_{SS}(\Delta S)^2 + \cdots .$$

The key point is

$$(\Delta S)^2 = \sigma^2 S^2\xi^2\Delta t + o(\Delta t).$$

Since $\mathbb{E}\xi^2 = 1$ and $\mathrm{Var}(\xi^2\Delta t) = 2(\Delta t)^2$, the term $\xi^2\Delta t$ is non-stochastic to first order and equals Δt in the limit. The second-order term is therefore $O(\Delta t)$ and cannot be dropped, unlike ordinary calculus. Hence

$$dG = \left(G_S\mu S + G_t + \tfrac{1}{2}G_{SS}\sigma^2S^2\right)dt + G_S\sigma S\,dz.$$

For $G = \log S$, $G_S = 1/S$, $G_{SS} = -1/S^2$, and $G_t = 0$, so

$$d\log S = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma\,dz.$$

Integrating over $[0,T]$ and using $W_T - W_0 \sim N(0,T)$ gives

$$\log S_T - \log S_0 \sim N\left((\mu - \tfrac{1}{2}\sigma^2)T,\ \sigma^2T\right),$$

which is the lognormal property (9th-ed. eqs. 14.17–14.19). The $-\sigma^2/2$ is the same correction derived heuristically in T5(b) and estimated in T8.

Checked example: Example 14.3 and Table 14.1

For $\mu = 0.15$, $\sigma = 0.30$, and one week $\Delta t = 0.0192$,

$$\frac{\Delta S}{S} = 0.15(0.0192) + 0.30\sqrt{0.0192}\,\xi = 0.00288 + 0.0415692\,\xi,$$

which the book rounds to $0.00288 + 0.0416\xi$. For the first two rows of Table 14.1,

$$\Delta S = 0.288 + 0.0415692(100)(0.52) = 2.4496 \approx 2.45 \quad (S = 100,\ \xi = 0.52),$$

$$\Delta S = 0.2941 + 0.0415692(102.45)(1.44) = 6.4277 \approx 6.43 \quad (S = 102.45,\ \xi = 1.44).$$

Both rows reproduce the table. The update is exactly A3's Euler scheme with $\sigma\sqrt{h}$, not the deliberately wrong σh.

Course connections and questions. T5(a)'s finite nonzero variance requires the $\sqrt{h}$ scaling; T6's moments follow from the lognormal law; T7's growth rate is $\mu - \sigma^2/2$; A2's exact sampler exponentiates the normal log price; A3's Euler step is (14.7)–(14.8). Hull does not treat estimation error, calibration, volatility clustering, or the positivity failure of Euler steps; those are the questions carried into A1–A3 and T8.

R1 reading notes, page 4 of 4: cross-book comparison and unresolved questions

Two books, two purposes. Tsay (pp. 2–23) is empirical and statistical: it starts from observed prices, defines returns, reports sample moments and distributional diagnostics, and asks what dependence is visible in a finite sample. Hull (Chapter 14) is structural and continuous-time: it starts from a Markov assumption and an SDE, derives the law of $\log S_T$ with Itô's lemma, and uses that law for analytic moments and simulation. The first is a toolkit for diagnosing data; the second is a model for propagating uncertainty. Neither, by itself, establishes weak-form efficiency or that a real price follows GBM.

Notation and a hidden parameter conflict. Tsay writes prices P_t, simple returns R_t, and log returns r_t; Hull writes S_t, μ, σ, and dz. The same symbol μ does not mean the same object. Hull's μ is the expected arithmetic rate of return in $dS/S = \mu\,dt + \sigma\,dz$; the annual log drift is $m = \mu - \sigma^2/2$. For observations spaced h years apart,

$$\mathbb{E}r_i = mh, \qquad \mathrm{Var}(r_i) = \sigma^2 h, \qquad \mathbb{E}(e^{r_i} - 1) = e^{\mu h} - 1.$$

T8 estimates m and σ from data and then reports $\widehat{\mu} = \widehat{m} + \widehat{\sigma}^2/2$. Setting $\widehat{\mu} = 252\bar{r}$ without the correction would conflate the two conventions. A1 uses $h = 1/252$; Tsay's data spacing plays the same role, but his discussion does not impose the constant-parameter law.

Assumptions and methods compared. Tsay's return aggregation is an identity, while his distributional statements are empirical descriptions; he does not require independent Gaussian increments. Hull explicitly assumes independent normal increments, constant μ and σ, and the Markov property, and he derives rather than assumes the lognormal terminal law. Tsay's tools are sample moments, histograms, Q–Q plots, and autocorrelations; Hull's are the SDE, Itô's lemma, analytic moments, and Monte Carlo paths. The methods answer different questions: A1's ACF and tail counts can reject a fitted i.i.d.-normal description but cannot identify the true law; A2's exact sampler removes discretization error only inside the assumed model. The two readings therefore complement each other: Tsay supplies the diagnostics, Hull supplies the model and the numerical target.

Unresolved questions. (i) How should dividend and split adjustments be mapped to the total-return interpretation of S_t? (ii) How much of A1's excess kurtosis is sampling noise, parameter instability, or genuine conditional heteroskedasticity? (iii) Hull's constant-parameter model gives no standard error for μ or σ; T8's plug-in standard error is a start, but model uncertainty remains. (iv) A fixed 95% prediction interval for S_T and a Monte Carlo confidence interval for $\mathbb{E}S_T$ answer different questions; only the latter shrinks with the replication count. (v) Euler's scheme can produce nonpositive multipliers, so positivity and discretization bias must be checked numerically rather than inferred from a smooth-looking path.

AI disclosure. The close reading, reconstructions, and arithmetic checks above were prepared with AI assistance from the public Tsay excerpt and the Hull Chapter 14 course copy described on page 3. All numbers were recomputed independently. The student should verify the cited locations and complete a personal reading before relying on these notes in an examination or declaring them as personal reading.